

ULC403: DESIGN AND ANALYSIS OF ALGORITHMS

L	T	P	Cr
3	0	2	4.0

Course Objectives: To provide students with the knowledge and skills necessary to design and analyze algorithms for solving computational problems.

Syllabus

Introduction and Complexity Analysis: Analysing algorithms, Complexity classes, Time and space trade-offs in algorithms, Recurrence relations, Analysis of iterative and recursive algorithms, Amortized Analysis.

Algorithm Design Techniques and Analysis

Divide and Conquer: Fundamentals of divide and conquer strategy, Applications such as maximum subarray problem, Strassen's algorithm for matrix multiplication, merge sort, quick sort etc.

Greedy Algorithms: Elements of greedy strategy, Applications such as activity selection, Huffman Coding, job sequencing, fractional knapsack problem, etc.

Dynamic Programming: Elements of dynamic programming, Memorization and tabulation approaches, Applications such as matrix multiplication, 0/1 knapsack, Longest common subsequence, Optimal binary search tree, etc.

Backtracking: Introduction, Applications such as N queen problem, sum of subsets, graph coloring, etc.

Branch and Bound Algorithm: General method, Applications such as 0/1 knapsack problem, Traveling salesperson problem etc.

Graph Algorithms: Introduction to graphs, Paths and Circuits, Euler Graphs, Hamiltonian graphs, Cut-sets, Connectivity and Separability, Covering and Partitioning, Strongly connected component, Topological sort, Max flow: Ford Fulkerson algorithm, max flow- min cut.

String Matching Algorithms: Suffix arrays, Rabin-Karp, Knuth-MorrisPratt (KMP), Boyer Moore algorithm.

Problem Classes: P, NP, NP-Hard and NP-complete, deterministic and non-deterministic polynomial time algorithm approximation, Randomized algorithms.

Laboratory Work

Implementation of various algorithmic techniques for solving common computational/engineering problems.

Course Learning Outcomes (CLOs) /Course Outcomes (COs):

On completion of this course, the students will be able to:

1. Analyse the complexity of algorithms and implement them in specific scenarios.
2. Apply common algorithmic techniques such as greedy, dynamic programming etc. to standard computational problems
3. Design solutions using appropriate data structures and algorithms such as string matching, randomized, approximation and graph algorithms.
4. Develop efficient algorithms for solving various computational challenging problems in computing

Text Books:

1. Cormen H. T., Leiserson E. C., Rivest L. R., and Stein C., Introduction to Algorithms, MIT Press (2009) 3rd ed.
2. Horwitz E., Sahni S., Rajasekaran S., Fundamentals of Computers Algorithms, Universities Press (2008) 2nd ed.

Reference Books:

1. Levitin A., Introduction to the design and analysis of algorithms, Pearson Education (2008) 2nd ed.
2. Aho A.V., Hopcraft J. E., Dulman J. D., The Design and Analysis of Computer Algorithms, Addison Wesley (1974) 1st ed.
3. Sedgewick R. and Wayne K., Algorithms, Addison-Wesley Professional (2011), 4th ed.

Evaluation Scheme

Evaluation Elements	Weightage %
Mid Semester Test (MST) and End Semester Examination (ESE)	70
Sessional may include assignments, regular lab assessments, quizzes, and minor projects (report, presentation etc.)	30